

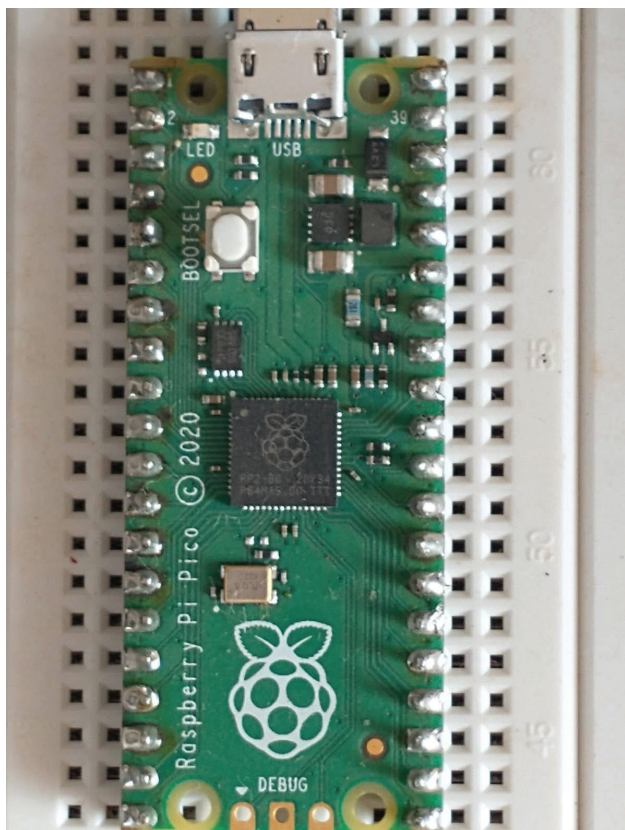


Figure 1: How the RPI Pico board looks

Now let's set up the software to program our Pico 1 board. First, download the necessary firmware for Pico from the MicroPython download page link provided in the References section at the end of this article. Next, connect the USB cable to your computer, and press and hold the *BOOTSEL* button

(visible in Figure 1) while inserting the cable into the micro USB port of the Pico board. You should see the Pico board mounted as a USB device in your computer. Just copy the downloaded UF2 firmware to the mounted Pico drive. The UF2 file will disappear from the mounted Pico drive once programming of the new firmware is complete. Now your Pico device will reset automatically and be ready for use.

Finally, we just need a Thonny Python IDE to interact with your Pico board, and program that for various IoT activities. You'll find Thonny already installed on your Raspberry Pi OS if you are using an RPI single board computer. There are multiple methods to install Thonny on various operating systems. I prefer the following command to set up Thonny in an isolated manner so that it does not interfere with an existing Python installation on a GNU/Linux system:

```
bash <(wget -O - https://thonny.org/installer-for-linux)
```

Just follow along and you'll have Thonny installed with its binaries and its uninstaller under the *./apps/thonny/bin/* directory path. The installer should also create a menu item in your desktop system. Otherwise, execute the command *apps/thonny/bin/thonny* to launch it. Thonny is intelligent enough to detect your Pico board is connected and ready to program, as shown in Figure 2. You can also configure Thonny through the 'Configure interpreter' menu item in the right bottom side of the IDE.

The Pico board has an LED (clearly visible in Figure 1) connected to its programmable input output pin 25. As indicated by the name itself, the Pico board has multiple programmable pins to interact with external or internal devices to build numerous IoT projects. Just click on *new*

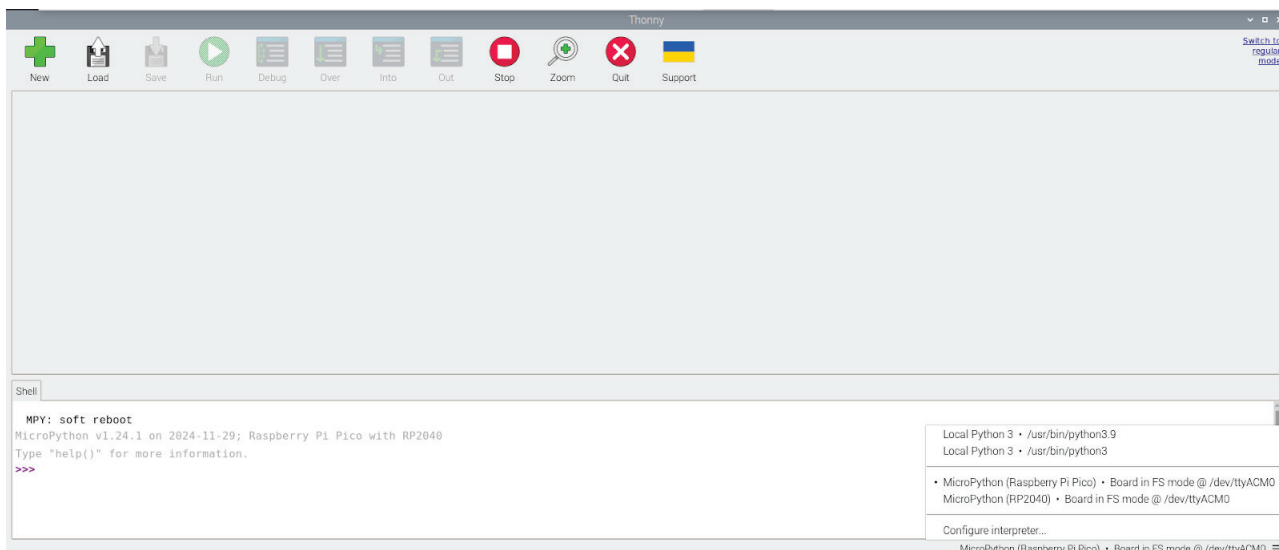


Figure 2. Thonny IDE GUI on launch